

MUHAMMAD ULUL ALBAB

Junior Software Engineer | Full-Stack Web & AI

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PROFILE

Junior Software Engineer with hands-on experience building full-stack web applications, AI-powered products, backend services, and APIs. Experienced with React, TypeScript/JavaScript, Node.js, Python/FastAPI, PostgreSQL, and AI APIs including Gemini. Built and deployed projects involving AI content generation, text-to-speech, FFmpeg media processing, vector search, and cloud services. Comfortable taking features from idea to implementation, debugging across the stack, and adapting quickly to unfamiliar technologies. Currently pursuing a Pharmacy degree while building a software engineering career through practical projects.

SKILLS & CORE TECH STACK

Languages: JavaScript, TypeScript, Python, PHP

Frontend: React, Vite, React Native, Expo, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, REST APIs

AI & Media: Gemini API, LLM Integration, RAG, SentenceTransformers, Text-to-Speech, FFmpeg

Database & Infrastructure: PostgreSQL, pgvector, Firebase Firestore, MySQL, Docker, Railway

Engineering: API Design, Authentication, State Management, Debugging, Modular Code, Git/GitHub

SELECTED PROJECTS

PodLearn AI — AI Podcast & Interactive Learning Platform *Node.js • Express • Gemini API • Edge TTS • FFmpeg • Supabase • Google OAuth*

- Built an end-to-end AI application that converts PDF/text learning materials into structured two-speaker podcast conversations with auto-generated quizzes.
- Integrated Gemini for content generation and Edge TTS for multi-speaker audio, with FFmpeg for server-side audio processing.
- Designed an interactive playback system with segment-by-segment audio playback and mid-podcast Q&A via a dedicated AI endpoint.
- Implemented Google OAuth authentication and Supabase-backed persistent podcast history, including delete functionality.
- Resolved deployment issues on Railway, including TTS request timeout tuning and Gemini API key rotation to handle API rate limits.

ScentDNA — AI Semantic Search & Recommendation Engine *FastAPI • PostgreSQL/pgvector • SentenceTransformers • Gemini • Docker*

- Built a semantic fragrance search engine that converts natural-language queries into vector embeddings and retrieves relevant products using PostgreSQL pgvector.
- Integrated Gemini to generate contextual recommendations from retrieved results using a retrieval-augmented generation workflow.
- Built a FastAPI REST API for search and recommendation endpoints with filtering and structured response schemas.
- Optimized model initialization and CPU resource usage for stable deployment in a constrained cloud container environment.

SchoolCom — School Management Platform *React Native • Expo • TypeScript • Firebase/Firestore*

- Developing a role-based school management platform covering attendance, assessments, student progress tracking, and administrative workflows.
- Implemented Firestore-backed data flows, role-aware navigation, and authorization guards across multiple user roles.
- Maintained type-safe, modular service and UI layers across an actively evolving multi-phase application.

Clinical Suite Dashboard *HTML5 • JavaScript • Tailwind CSS*

- Built a clinical management dashboard with structured data workflows and automated clinical parameter calculators using reusable JavaScript components and responsive UI patterns.

LDR Anchor — Mobile Application *React Native • Firebase Firestore • FCM V1*

- Built an interactive Android application for long-distance couples featuring real-time mood synchronization, virtual hugs, and automated push notifications via Firebase Cloud Messaging (FCM V1 API).

EDUCATION

Universitas Pakuan Bogor, Indonesia — Bachelor's of Pharmacy Candidate | [2023 – Present]